Problem 15: Fairness-Aware Recommender System with Cold-Start Handling and Off-Policy Evaluation

Domain: Recommender Systems | Causal ML | Bandits | Responsible AI

Difficulty: Hard

Status: NEW (added to reach 15)

Problem

Production recommenders are trained on logged interactions that reflect the previous system's choices (exposure and popularity bias). They struggle with new users and items, can under-expose some creators or groups, and are usually evaluated with offline metrics that do not predict online performance.

Objective

Build a recommender that handles cold start, balances relevance with fairness/diversity, learns online with bandits, and is evaluated with unbiased off-policy methods.

Users

Product/ML teams at marketplaces, media, e-commerce and learning platforms; end users; item providers.

Core Requirements

- Ingest user-item interaction data with timestamps.
- Baselines: popularity, item-kNN, matrix factorization (ALS/BPR).
- Neural or two-tower retrieval model plus a ranking model.
- Cold-start: content features for new items, onboarding strategy for new users.
- Time-based train/validation/test split.
- Fairness/diversity re-ranking (e.g., provider exposure parity, long-tail coverage) with a tunable relevance-fairness trade-off.
- Contextual bandit module (e.g., LinUCB / Thompson sampling) for exploration.
- Offline policy evaluation with IPS / doubly robust estimators on logged bandit feedback.
- Explanations for recommendations ("because you…").
- Dashboard: metrics, fairness exposure, A/B-style simulation results.

Advanced Challenges

- Popularity and position bias correction.
- Building a user simulator to test long-term effects (feedback loops, filter bubbles).
- Multi-objective optimization (relevance, diversity, fairness, revenue).
- Counterfactual evaluation validity: when IPS variance explodes.
- Scalable retrieval with ANN indexes (FAISS) and latency budget.
- Privacy: avoid using sensitive attributes directly; audit proxies.

Suggested Datasets

MovieLens (1M/20M), Amazon Reviews 2023, Yelp, KuaiRec (fully-observed matrix, useful for unbiased eval), Open Bandit Dataset (logged bandit feedback with propensities).

Suggested Tech

Python, PyTorch, implicit / LightFM / RecBole, FAISS, Open Bandit Pipeline, scikit-learn, FastAPI, Streamlit.

Reference Architecture

Event log → feature store → candidate generation (two-tower + ANN) → ranker → fairness/diversity re-ranker → bandit exploration layer → serving API → logging with propensities → off-policy evaluator → dashboard.

Evaluation

Recall@K, nDCG@K, MAP; cold-start-only metrics; coverage, Gini of exposure, provider-exposure disparity; IPS/DR value estimates with confidence intervals; cumulative reward in

simulation; p95 latency.

Responsible AI Note

State which fairness definition you chose and why; report the accuracy cost of fairness constraints honestly.

Pitfalls

Random split leaking future interactions; evaluating only on popular items; logging policy without propensities, making off-policy evaluation impossible.

Suggested Timeline (48h)

0-8h data + baselines; 8-20h two-tower + cold-start; 20-30h fairness re-ranking; 30-40h bandits + off-policy eval; 40-48h dashboard.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work